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NN ASSINGMENT 1

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1 Part 1: Regression Problems

1.1 Problem Summary and Required Tasks

In this part we apply polynomial regression to two real-world datasets and study how outliers affect model performance.

Problem 1 — Insurance Data. The number of insured persons is recorded for the years 1987–1996. We are required to:

1. Fit linear, quadratic, and cubic polynomials and compare R^2 values to determine the best-fit model.
2. Perform hand analysis (Normal Equation) for the linear case.
3. Detect and remove outliers, then refit and compare results.
4. Graph the best-fit function together with the scatter plot.
5. Predict the average number of insured persons in 1997.

Problem 2 — Heating Oil. Oil consumption data is given with two features (temperature, insulation). We are required to:

1. Fit linear and quadratic models and compare R^2 values.
2. Detect and remove outliers using the Mahalanobis distance, then refit and compare.
3. Predict oil usage for a temperature of 15°F and insulation of 5 inches.

1.2 Implementation Flow

The following steps are performed for both problems:

1. Load the dataset and define the feature variables.
2. Fit polynomial regression models using the Normal Equation.
3. Evaluate each model by computing R^2 and comparing fits.
4. Detect outliers (Z-score for univariate data, Mahalanobis distance for multivariate data).
5. Remove the identified outlier and refit all models on the cleaned data.
6. Compare performance before and after outlier removal.
7. Use the best-fit model to make predictions on unseen inputs.
8. Generate all relevant plots (scatter, overlay, surface, prediction).

1.3 Problem 1 — Insurance Data Polynomial Regression

The numbers of insured persons y with an insurance company for the years 1987 to 1996 are shown in the table below. We let x denote the number of years since 1987 (i.e. $x = 0$ for 1987, $x = 9$ for 1996).

Year	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
x	0	1	2	3	4	5	6	7	8	9
y	12400	10900	10000	1050	9500	8900	8000	7800	7600	7200

Table 1: Insurance data.

1.3.1 (a.1) Scatter Plot

The data is plotted in a 2-D scatter plot to visualise the overall trend.

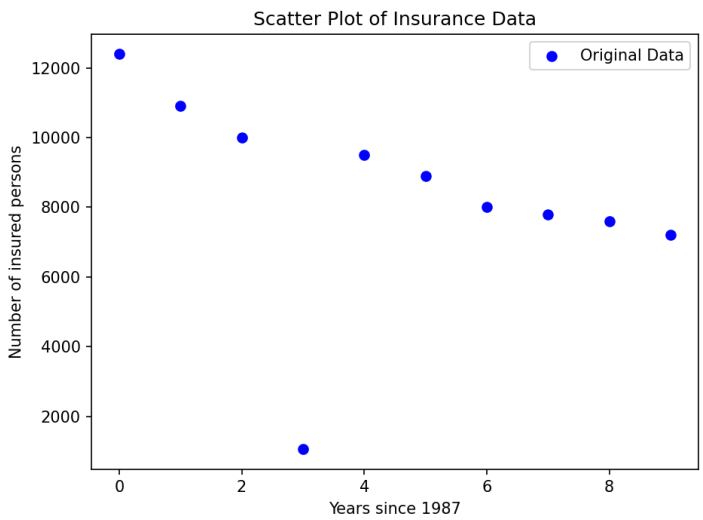

Figure 1: Scatter plot of the insurance data.

Comment. As shown in the figure above, the number of insured persons generally decreases over the years. However, in 1990 ($x = 3$) the recorded value drops to 1 050 — far below the neighbouring values — which strongly suggests a data-entry error. This point will likely distort any regression model and should be treated as an outlier.

1.3.2 (a.2) Hand Analysis — Linear Fit (on cleaned data)

We first carry out the linear regression *by hand* using the Normal Equation. The outlier at $x = 3$ is removed first, leaving $n = 9$ data points with $x \in \{0, 1, 2, 4, 5, 6, 7, 8, 9\}$.

The linear model is $y = \theta_0 + \theta_1 x$. The design matrix and target vector are:

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 6 \\ 1 & 7 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 12400 \\ 10900 \\ 10000 \\ 9500 \\ 8900 \\ 8000 \\ 7800 \\ 7600 \\ 7200 \end{pmatrix}$$

Step 1 — Compute $\mathbf{X}^T\mathbf{X}$:

$$\mathbf{X}^T\mathbf{X} = \begin{pmatrix} \sum 1 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 9 & 42 \\ 42 & 276 \end{pmatrix}$$

Step 2 — Compute the inverse $(\mathbf{X}^T\mathbf{X})^{-1}$:

$$\det = 9 \times 276 - 42 \times 42 = 2484 - 1764 = 720$$

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{720} \begin{pmatrix} 276 & -42 \\ -42 & 9 \end{pmatrix} = \begin{pmatrix} 0.3833 & -0.0583 \\ -0.0583 & 0.0125 \end{pmatrix}$$

Step 3 — Compute $\mathbf{X}^T\mathbf{y}$:

$$\mathbf{X}^T\mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} 82\,300 \\ 341\,600 \end{pmatrix}$$

Step 4 — Solve for $\boldsymbol{\theta}$:

$$\boldsymbol{\theta} = (\mathbf{X}^T\mathbf{X})^{-1}\,\mathbf{X}^T\mathbf{y} = \begin{pmatrix} 0.3833 & -0.0583 \\ -0.0583 & 0.0125 \end{pmatrix} \begin{pmatrix} 82\,300 \\ 341\,600 \end{pmatrix} = \begin{pmatrix} 11\,621.6667 \\ -530.8333 \end{pmatrix}$$

Thus the hand-calculated linear equation is:

$$\boxed{y = -530.83\,x + 11\,621.67}$$

Step 5 — Compute R^2:

$$SS_{\text{res}} = \sum (y_i - \hat{y}_i)^2 = 1\,339\,500.00 \qquad SS_{\text{tot}} = \sum (y_i - \bar{y})^2 = 23\,882\,222.22$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{1\,339\,500}{23\,882\,222} = \boxed{0.9439}$$

Comment. The hand-calculated results agree perfectly with the values produced by our Python code, which confirms the correctness of both approaches. The linear model already captures about 94% of the variance in the cleaned data, indicating a strong linear trend.

1.3.3 (a.3) Polynomial Fitting — With Outlier

We aim to fit polynomial models (linear, quadratic, cubic) to the given data points. The general form is

$$y = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $\hat{y}$ is the predicted value. The parameters $\boldsymbol{\theta}$ are estimated using the Normal Equation:

$$\boldsymbol{\theta} = \left(\mathbf{X}^T \mathbf{X}\right)^{-1} \mathbf{X}^T \mathbf{y}$$

where $\mathbf{X}$ is the design matrix (expanded with polynomial terms of x), and $\mathbf{y}$ is the vector of observed outputs.

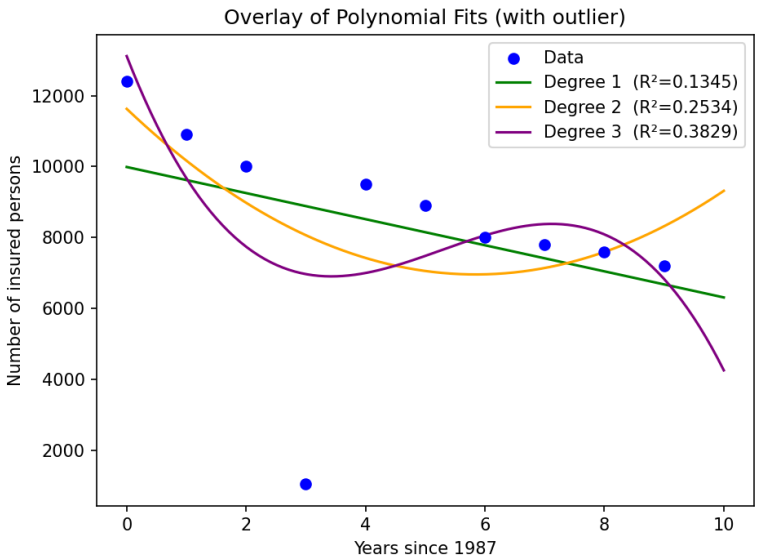

Figure 2: Overlay of linear, quadratic, and cubic fits (with outlier).

Degree	R^2	**% increase from previous**
Linear ($n = 1$)	0.1345	—
Quadratic ($n = 2$)	0.2534	88.3%
Cubic ($n = 3$)	0.3829	51.1%

Table 2: R^2 comparison with outlier present.

Comment. With the outlier still included, all three models struggle to explain the data — the best R^2 is only 0.3829 for the cubic fit. The large percentage increase from linear to quadratic (88.3%) suggests that some curvature helps, but overall the outlier is pulling every model away from the true trend. This is a clear sign that the suspicious data point at $x = 3$ needs to be addressed before drawing conclusions.

1.3.4 (b) Outlier Detection and Removal

From the scatter plot, the value at $x = 3$ ($y = 1050$) stands out immediately. To verify this quantitatively, we compute the mean $\bar{y}$ and standard deviation s of all 10 observations, then calculate the standardised score for each point:

$$Z_i = \frac{y_i - \bar{y}}{s}$$

Any observation whose $|Z|$ exceeds 2 is considered statistically anomalous.

Running this calculation, the point at $x = 3$ receives $Z = -2.65$, while every other observation stays within $|Z| < 1.3$. This single extreme score confirms what the plot already suggested: the 1990 reading is a genuine outlier and should be excluded before fitting. With the outlier removed, $n = 9$ data points remain.

1.3.5 (b) Polynomial Fitting — After Outlier Removal

After removing the outlier, all three polynomial models are re-estimated on the cleaned 9-point dataset.

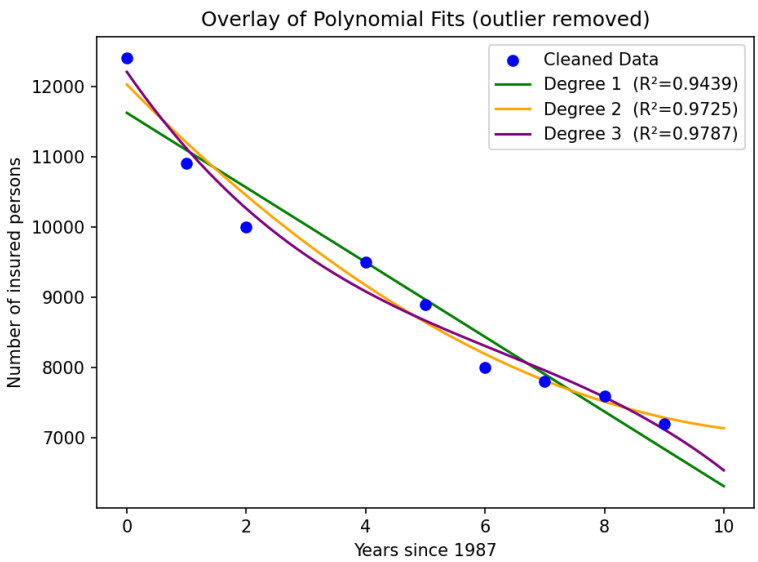

Figure 3: Overlay of polynomial fits after outlier removal.

Comment. After removing the outlier, the three curves are much closer together and all

follow the natural downward trend of the data. Visually, even the linear fit captures the pattern well.

Degree	R^2 (with outlier)	R^2 (cleaned)	% increase (cleaned)	Improvement
Linear	0.1345	0.9439	—	+0.8094
Quadratic	0.2534	0.9725	3.03%	+0.7191
Cubic	0.3829	0.9787	0.64%	+0.5958

Table 3: R^2 comparison before and after outlier removal.

Comment. Once the outlier is excluded, every model's R^2 improves substantially. The linear fit jumps from 0.1345 all the way to 0.9439, showing that the single bad data point was responsible for most of the earlier poor performance. Looking at the cleaned results, the step from linear to quadratic adds only 3.03% to R^2, and quadratic to cubic adds a mere 0.64%. Both increments are negligible compared to the complexity they introduce. Therefore the **linear model** is the most sensible choice here — it already explains over 94% of the variance with just two parameters.

The fitted equations on the cleaned data are:

$$\begin{aligned}
\text{Linear:} \quad & y = -530.83x + 11\,621.67 && R^2 = 0.9439 \\
\text{Quadratic:} \quad & y = 37.44x^2 - 863.13x + 12\,024.17 && R^2 = 0.9725 \\
\text{Cubic:} \quad & y = -7.46x^3 + 140.17x^2 - 1221.74x + 12\,204.42 && R^2 = 0.9787
\end{aligned}$$

1.3.6 (c) Graph of Best-Fit Function

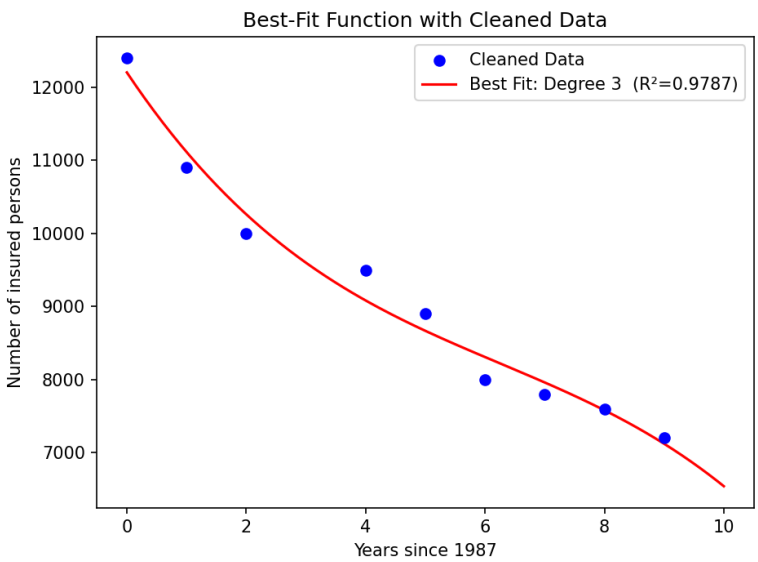

Figure 4: Best-fit polynomial (linear) plotted with the cleaned data.

Comment. The figure shows the linear model plotted alongside the cleaned data points. The fit closely follows the downward trend, confirming that the linear model is indeed a good representation of the underlying relationship.

1.3.7 (d) Prediction for 1997

Using the regression models, we predict the average number of insured persons in 1997 ($x = 10$). We report predictions from all three models both with and without the outlier, to show its impact on the forecasts.

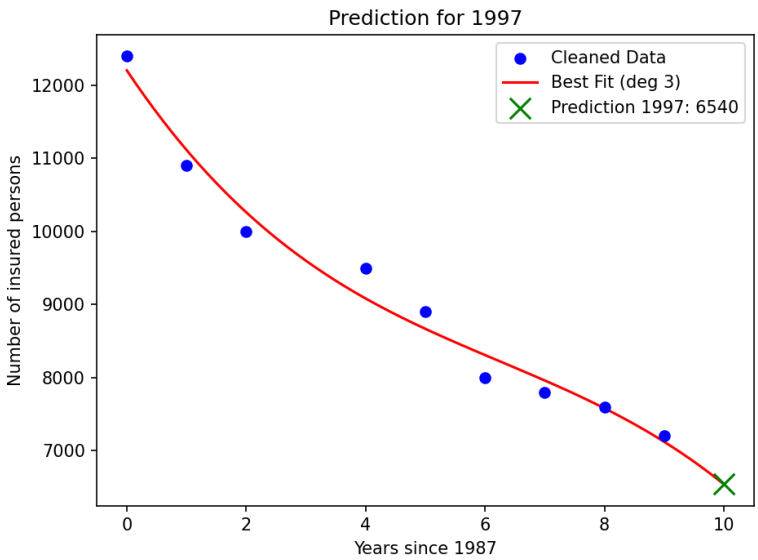

Figure 5: Prediction for 1997 using the best-fit model on cleaned data.

Degree	Prediction (with outlier)	Prediction (cleaned)
Linear	6 313	6 313
Quadratic	9 318	7 137
Cubic	4 260	6 540

Table 4: Predicted number of insured persons in 1997.

Comment. Using the selected linear model on the cleaned data, we get a forecast of roughly **6 313** insured persons for 1997. Notice that the linear prediction happens to be identical with and without the outlier — the single bad point did not shift the overall slope enough to matter at $x = 10$. The quadratic and cubic predictions, on the other hand, diverge noticeably between the two scenarios, which is a reminder that higher-degree polynomials amplify the effect of noisy data, especially when extrapolating beyond the training range.

1.4 Problem 2 — Heating Oil Multiple Regression

The goal is to develop a model for estimating heating oil used for a single-family home in January, based on average temperature and amount of insulation in inches. The dataset contains 15 observations.

Oil (gal)	Temp (°F)	Insulation (in)
270	40	4
362	27	4
162	40	10
45	73	6
91	65	7
233	65	40
372	10	6
305	9	10
234	24	10
122	65	4
25	66	10
210	41	6
450	22	4
325	40	4
52	60	10

Table 5: Heating oil dataset.

1.4.1 (a.1) Linear Model

Let $f(\mathbf{x}, \boldsymbol{\theta})$ be the output function of the linear model, where the input consists of two features: x_1 (average temperature) and x_2 (insulation thickness). The model is:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

Parameters are estimated using least-squares regression (Normal Equation).

The fitted equation is:

$$\hat{y} = 426.97 - 5.27\,x_1 + 1.97\,x_2 \qquad (R^2 = 0.6967)$$

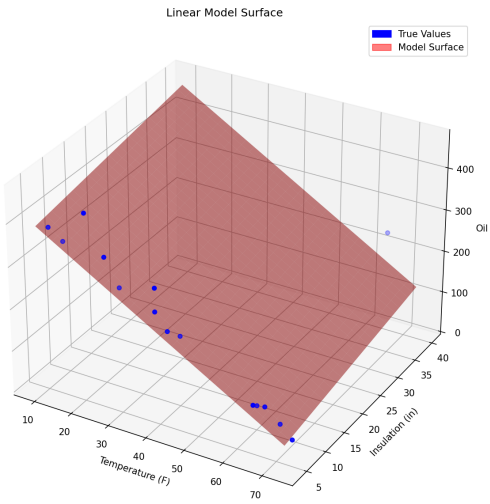

Figure 6: Linear model regression surface vs. true values.

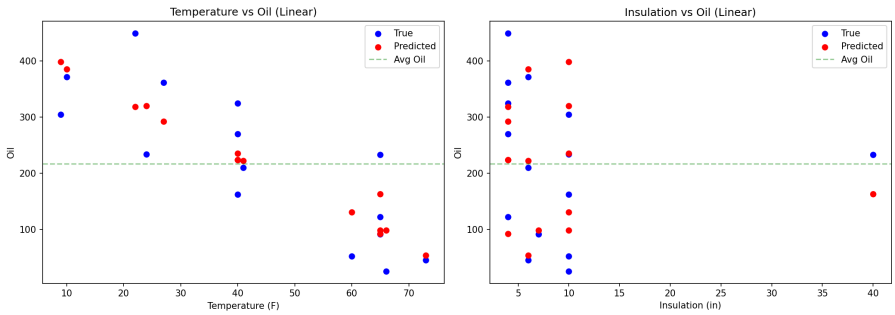

Figure 7: 2D projections: Temperature vs Oil and Insulation vs Oil (Linear).

Comment. The linear model yields $R^2 = 0.6967$, meaning it explains about 70% of the variance. As we can see from the 2D projections, the linear fit follows the dominant trend — oil consumption decreases as temperature rises — but the insulation coefficient is positive ($+1.97$), which contradicts physical intuition. This is likely caused by the outlier (Insulation = 40) pulling the coefficient upward.

1.4.2 (a.2) Quadratic Model

The quadratic model includes all second-order terms and the cross-product:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1 x_2 + \theta_5 x_2^2$$

The fitted equation is:

$$\hat{y} = 680.03 - 6.17\,x_1 - 41.72\,x_2 - 0.01\,x_1^2 + 0.24\,x_1 x_2 + 0.66\,x_2^2 \qquad (R^2 = 0.9742)$$

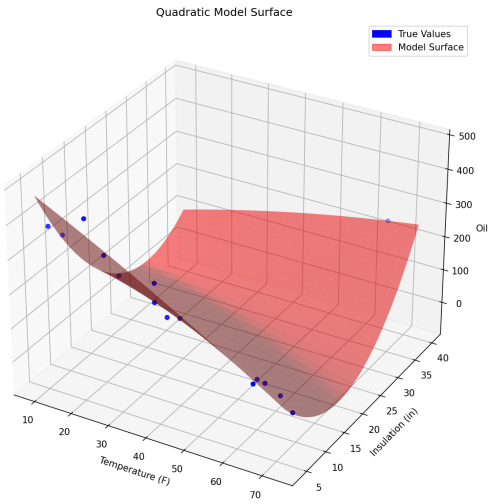

Figure 8: Quadratic model regression surface vs. true values.

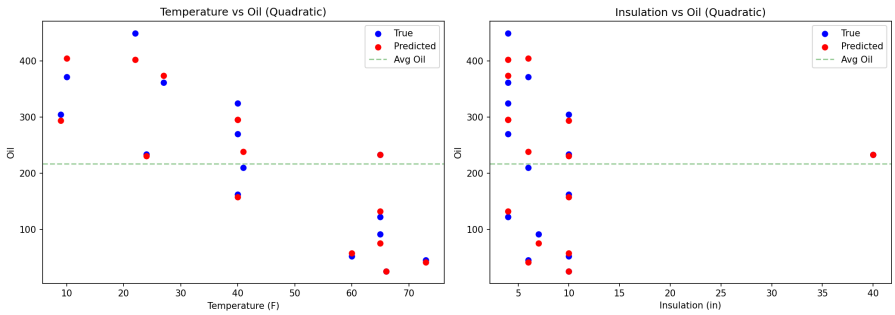

Figure 9: 2D projections: Temperature vs Oil and Insulation vs Oil (Quadratic).

Comment. The quadratic model achieves $R^2 = 0.9742$, a substantial jump from the linear model. It can capture the non-linear interactions between temperature and insulation and handles the high-insulation data point more gracefully.

R^2 **Comparison:**

Model	R^2
Linear	0.6967
Quadratic	0.9742

Table 6: R^2 comparison for linear and quadratic models (all data).

Comment. As shown in the table, the quadratic model provides a significantly better fit than the linear one. It explains 97.4% of the variance in the data compared to only 69.7% for the linear model.

1.4.3 (b) Outlier Detection and Removal

Since the input space is two-dimensional (temperature and insulation), a simple Z-score on individual features would miss correlations between them. Instead, we use the **Mahalanobis distance**, which takes the full covariance structure into account:

$$D_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{S}^{-1} (\mathbf{x} - \boldsymbol{\mu})}$$

We compare each point's D_M against a χ^2-based threshold at the 95% confidence level ($p = 2$ features). Any point that exceeds this threshold is flagged.

Comment. The only flagged observation is the one with Temp = 65°F and Insulation = 40. Looking at the raw data, all other insulation values fall between 4 and 10, so a value of 40 is clearly anomalous. Interestingly, this point was also the reason the linear model's insulation coefficient came out positive ($+1.97$) earlier — once we remove it, the coefficient flips to a physically meaningful negative value.

1.4.4 (b.1) Linear Model after Removing Outlier

After removing the outlier, the linear model is refitted on the 14-point cleaned dataset.

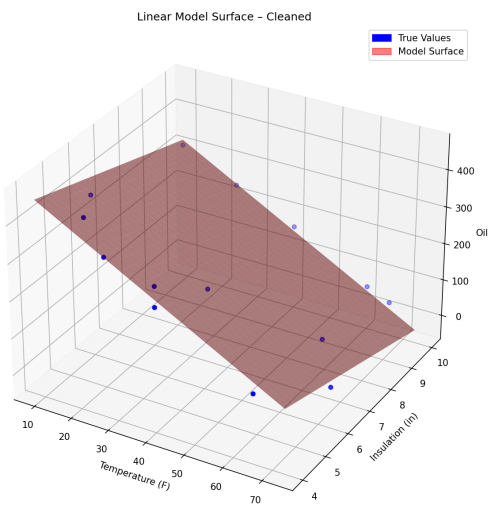

Figure 10: Linear model surface after outlier removal.

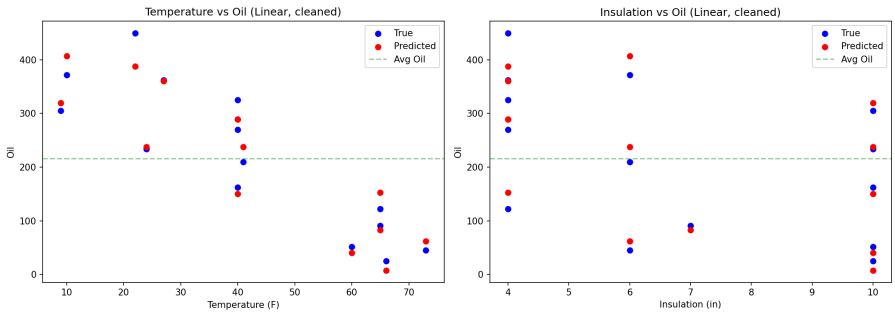

Figure 11: 2D projections after outlier removal (Linear).

Comment. After removing the outlier, the linear model's R^2 jumps from 0.6967 to 0.9606 — a dramatic improvement. The insulation coefficient is now negative (-23.23), which makes physical sense: more insulation means less heat loss, hence less oil consumption.

1.4.5 (b.2) Quadratic Model after Removing Outlier

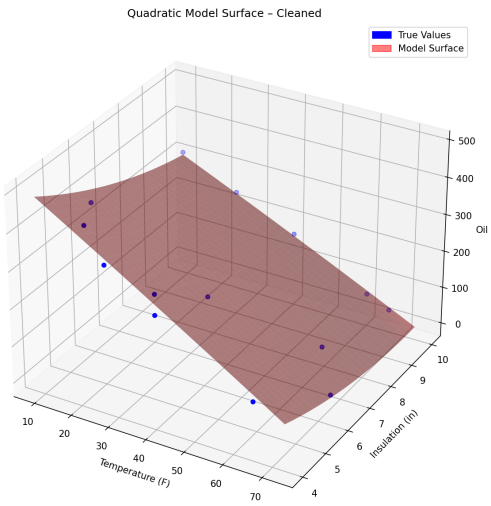

Figure 12: Quadratic model surface after outlier removal.

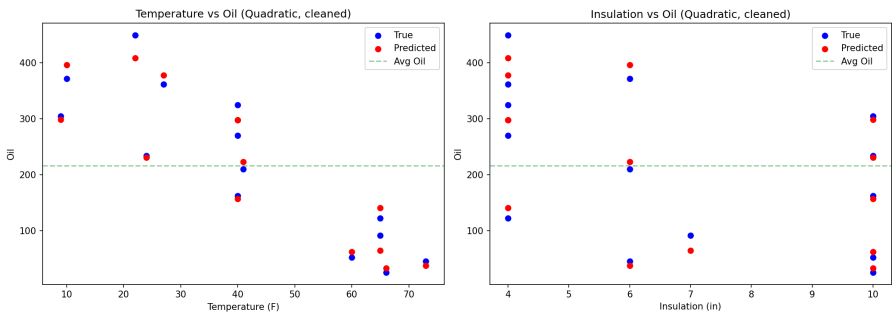

Figure 13: 2D projections after outlier removal (Quadratic).

Comparison of Performance Before and After Removing Outlier:

Model Type	R^2 (before removing outlier)	R^2 (after removing outlier)	Improvement
Linear	0.6967	0.9606	+0.2639
Quadratic	0.9742	0.9769	+0.0027

Table 7: R^2 before and after outlier removal for Problem 2.

Comment. The table reveals an interesting asymmetry: the linear model benefits enormously from outlier removal (R^2 rises by 0.2639), whereas the quadratic model barely changes ($+0.0027$). This tells us that the extra flexibility of the quadratic terms was already compensating for the outlier internally, while the linear model had no such mechanism.

Once the data is clean, however, both models perform almost equally well — the gap between them shrinks to just 1.63%. At that point, the added complexity of five quadratic coefficients is hard to justify. We therefore select the **linear model** as our preferred fit for the cleaned dataset, since it achieves $R^2 = 0.9606$ with only three parameters.

The cleaned-data equations are:

$$\text{Linear:} \quad \hat{y} = 601.60 - 5.48\,x_1 - 23.23\,x_2 \qquad R^2 = 0.9606$$

$$\text{Quadratic:} \quad \hat{y} = 785.52 - 6.96\,x_1 - 72.16\,x_2 - 0.004\,x_1^2 + 0.26\,x_1x_2 + 2.74\,x_2^2 \quad R^2 = 0.9769$$

1.4.6 (c) Prediction for Temp = 15°F, Insulation = 5 in

Using the regression models fitted on the cleaned data, we predict the oil consumption for an average temperature of 15°F and insulation of 5 inches:

Model (cleaned)	Predicted Oil (gal)
Linear	403.22
Quadratic	407.35

Table 8: Predicted oil usage for Temp = 15°F and Insulation = 5 in.

Comment. The two predictions are very close (about 400–410 gallons), which is expected for such cold weather with thin insulation. The fact that both models agree on the forecast reinforces our earlier observation: on clean data, the quadratic terms add almost no predictive value over the simpler linear model.

2 Part 2: Classical ML Modeling

2.1 Problem Summary and Required Tasks

ReducedMNIST is used with 10 classes (digits 0–9), where the training split has 1000 samples per class and the testing split has 200 samples per class.

The required tasks are:

1. Extract three feature families for train and test sets:
   - DCT with 225 dimensions,
   - PCA with cumulative variance at least 95%,
   - HOG features.
2. Train and evaluate:
   - K-means per class with $K = 1, 4, 16, 32$,
   - SVM with linear and nonlinear kernels.
3. Report comparative accuracy and processing time in one final table.
4. Show confusion matrices only for the best K-means result and the best SVM result.
5. Add conclusions.

2.2 Implementation Flow

The implemented flow in Part_2 scripts is:

1. Load ReducedMNIST (JPG folder dataset via KaggleHub, with safe fallbacks).
2. Normalize pixel values to $[0, 1]$.
3. Extract DCT, PCA, and HOG features.
4. Train K-means per class for $K \in \{1, 4, 16, 32\}$ and evaluate on test data.
5. Train SVM with linear and RBF kernels and evaluate on test data.
6. Store predictions, measure real elapsed time, and print summary tables.
7. Visualize confusion matrices and comparative charts.

2.3 Nonlinear SVM Used (RBF) and Its Specifications

The nonlinear model used in this implementation is **C-SVC with RBF kernel** from scikit-learn.

The kernel function is:

$$K(\mathbf{x}, \mathbf{z}) = \exp\left(-\gamma \|\mathbf{x} - \mathbf{z}\|^2\right)$$

which enables curved (nonlinear) decision boundaries for classes that are not linearly separable.

Exact settings used in code (Part_2/step3_svm.py):

- `kernel='rbf'`
- $C = 10.0$
- $\gamma =$ `scale`
- `decision_function_shape='ovo'` (one-vs-one multiclass)

Feature scaling step before SVM training/inference:

- `StandardScaler()` is fitted on training features.
- The same scaler is applied to both train and test features.

For $\gamma =$ `scale`, scikit-learn uses:

$$\gamma = \frac{1}{n_{\text{features}} \operatorname{Var}(X)}$$

All remaining `SVC` parameters are left at library defaults.

2.4 Comparative Results (Accuracy and Processing Time)

Table 9 summarizes all required combinations with measured real elapsed time.

Classifier	**DCT**		**PCA**		**HOG**	
	Acc. (%)	Time (s)	Acc. (%)	Time (s)	Acc. (%)	Time (s)
K-means $K = 1$	86.85	0.01	86.80	0.00	91.35	0.04
K-means $K = 4$	92.65	2.19	92.30	0.61	93.60	8.00
K-means $K = 16$	94.85	1.50	94.80	1.51	95.00	17.85
K-means $K = 32$	96.15	2.05	95.70	2.14	96.00	29.79
SVM Linear	92.60	1.78	90.65	2.66	96.70	10.80
SVM Nonlinear (RBF, $C = 10$, $\gamma =$scale)	96.90	5.05	95.40	12.54	96.70	31.02

Table 9: Part 2 required comparative table: accuracy and processing time.

2.5 Best Results and Confusion Matrices

Best K-means result: **K-means** $K = 32$ **with DCT** (96.15%).
Best SVM result: **SVM Nonlinear (RBF) with DCT** (96.90%).

Insert confusion matrix image for best K-means result here (K=32 with DCT).

Figure 14: Confusion matrix for the best K-means model.

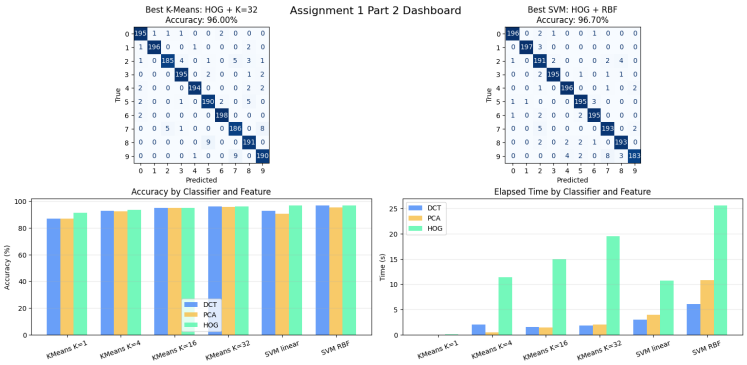

Figure 15: Confusion matrix for the best SVM model.

2.6 Conclusions

1. Increasing K in class-wise K-means consistently improves performance, with best result at $K = 32$.

2. Nonlinear SVM (RBF) gives the highest overall accuracy among all tested combinations.

3. HOG is competitive, especially for linear SVM, but incurs higher runtime.

4. DCT features produced the strongest best-case accuracy for both classifier families in this run.

5. Reporting both accuracy and elapsed time is important because the best-accuracy model is not always the fastest one.